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A novel machine learning approach for assessing water quality using *Daphnia* swimming behavior

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Abstract
To bridge the gap between complex water chemistry data and public understanding, we developed a novel *Daphnia magna* biomonitoring approach combined with deep learning to assess water quality. We recorded *Daphnia* swimming behavior under different salinity levels (0, 0.01, 0.1, 1, 10%) and extracted key movement features. A Multi-Layer Perceptron (MLP) model was then trained on these features to classify water quality levels. The MLP achieved an overall accuracy of ~ 80%, with class-wise precision, recall, and F1-scores generally ranging from ~ 0.7 to 0.9. Behavioral analysis showed that *Daphnia*'s average swimming speed and acceleration decreased at higher salinities ($p < 0.01$), indicating sensitivity to pollutant stress. This study is the first to integrate video-based *Daphnia* monitoring with machine learning for water quality assessment, demonstrating a simple, low-cost method to empower citizen scientists and enhance environmental education.

Keywords *Daphnia magna*, Swimming behavior, Water quality, Deep learning, Citizen science, Environmental education, Biomonitoring, Salinity toxicity

1 Introduction

The sustainable management of water resources and ecosystems has long been a key issue in global environmental protection and sustainable development. With population growth, increased urbanization, and the discharge of various industrial and agricultural wastewater, water pollution not only affects aquatic biodiversity but also threatens human drinking water safety. In recent years, "citizen science" has gradually emerged, emphasizing the participation of ordinary people in scientific research, which increases data sources and also enhances environmental awareness [1, 2]. Citizen science involves the participation of "non-scientists" in traditional scientific activities—for example, volunteers can contribute by collecting samples, making observations, identifying images, and/or sharing computing power for projects [3].

The World Water Monitoring Day (WWMD) experience is a global public water education activity that incorporates elements of citizen science. WWMD participants measure basic water quality indicators such as temperature, pH, dissolved oxygen, and

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turbidity using simple kits. By participating in WWMD, the public is expected to (1) enhance their understanding of clean water issues; (2) encourage innovation and technology exchange; (3) rekindle public interest in water stewardship; and (4) promote individual management of sustainable water resources [4]. These activities are very popular and successfully attract people to engage with local water bodies in daily life. However, participants often find it difficult to interpret what the measured water quality data implies for aquatic ecosystems or human health. This is thought to be due to the weak linkage between water quality numbers and people's daily experiences. Indeed, studies show that the public lacks sufficient knowledge of water quality parameters and their significance [5].

In the field of water quality monitoring, traditional methods rely on chemical analyses (e.g., measuring pH, dissolved oxygen, turbidity, ammonia nitrogen), which provide quick quantitative results for specific parameters but often lack real-time feedback on biological toxicity or ecological effects. Therefore, biomonitoring methods that are both ecologically sensitive and low in operational barriers have gained attention. The swimming behavior of *Daphnia*—which is complex and multi-parametric—is considered one of the most sensitive biomarkers for water toxicity [6]. *Daphnia* are a focus in this regard because they exhibit high sensitivity to various pollutants (including heavy metals, pesticides, and microplastics) and can show abnormal behavior even before water quality degradation leads to mass mortality [7, 8]. Among behavioral indicators, swimming patterns (such as speed and acceleration have recently gained particular favor for detecting sub-lethal contaminant effects,many studies have noted that *Daphnia*'s speed, acceleration, and trajectory characteristics change significantly under toxic stress [8–10].

However, there are still challenges to popularizing "*Daphnia* swimming behavior monitoring" in environmental education and public participation contexts. First, traditional observation of *Daphnia* behavior requires expensive laboratory equipment and long, labor-intensive manual recording. Second, the volume of trajectory or speed data collected is large, and the interactions among parameters are complex and not intuitively understandable, often requiring advanced statistical or machine learning techniques to interpret effectively [11]. In view of these issues, this study introduces a deep learning approach combined with simple smartphone video recording to design a straightforward *Daphnia* behavior monitoring process that can be completed quickly and evaluates its feasibility in citizen science and environmental education activities. It is hoped that through this method, ordinary people can rapidly collect biologically relevant water quality data using smartphones or other simple devices and subsequently analyze the data with a cloud-based or local machine learning model, thereby expanding the scale and efficiency of water resource monitoring [12–14]. To our knowledge, this study is the first to integrate video-based *Daphnia* monitoring with a deep learning model for accessible water quality assessment in an educational context. The purpose of this study is to examine whether image analysis of *Daphnia* swimming behavior combined with a deep learning model can effectively distinguish different levels of pollution and to assess the application potential of this method in citizen science, thereby empowering participants and enhancing environmental protection behaviors.

2 Materials and methods

2.1 Experimental materials and culture

In this study, *Daphnia magna* individuals were obtained from a long-term, stable laboratory culture. Previous studies have shown that *Daphnia magna* is highly sensitive to a broad range of pollutants (e.g., pesticides, heavy metals, nanoparticles) and is easy to culture in large numbers [7, 9]. Maintaining stable breeding conditions enables *Daphnia* to produce many offspring with similar genetic backgrounds (via clonal reproduction), thereby reducing experimental variation. Our culture conditions followed the standard protocol of DAPHTOXKIT F magna (developed by G. Persoone, Ghent University, Belgium), which uses an artificial freshwater formulation: for 1 L of deionized water, add 64.75 mg $NaHCO_3$, 294 mg $CaSO_4{\cdot}2H_2O$, 123.25 mg $MgSO_4$, and 5.75 mg KCl. This provides essential minerals to meet *Daphnia*'s physiological needs. *Daphnia* were reared in a 4 L transparent glass aquarium (with ~ 2 L of actual water volume) and fed daily with a diet of algae or yeast prepared according to Taiwan EPA standards. The culture environment was maintained at 20–25 ℃ with a 16:8 h light:dark cycle, simulating standard laboratory conditions. Once the *Daphnia* reached an adequate size (~ 2–3 mm), they were deemed ready for use in subsequent tests.

2.2 Daphnia swimming behavior testing and trajectory analysis

For the experiments, we used salinity (NaCl concentration) as a representative pollutant stress to evaluate *Daphnia*'s movement patterns under different stress levels. Prior to exposure, healthy adult *Daphnia* were collected from the stock culture and transferred to a 12-well plate. In each well, 1 mL of double-strength (2 ×) culture medium was mixed with 1 mL of a 2 × NaCl solution to achieve the desired final salinity levels of 0%, 0.01%, 0.1%, 1%, or 10%. Each treatment (salinity level) was replicated in three wells. During the 48-h exposure period, *Daphnia* were not fed, in order to simulate outdoor conditions (where food may be limited) and to avoid confounding influences of feeding [14, 15].

For recording behavior, a smartphone camera (mounted on a fixed tripod) was positioned approximately 12 cm above the well plate. A small portable LED flashlight, diffused through paper, was placed beneath the plate to provide even backlighting and minimize phototactic bias in *Daphnia* movements [11, 16]. Each video recording lasted about 30 s and was saved as an MP4 file. We then used Tracker motion analysis software (Open Source Physics) to extract the x–y coordinates of the *Daphnia* in each frame over time. From these coordinates and time data, we calculated movement parameters including instantaneous velocity (v), acceleration (a), kinetic energy (reported as momentum p, since mass is constant), and angular speed (ω). Because the wells were shallow (~ 5 mm depth), vertical movement was minimal and the motion could be treated as essentially two-dimensional, simplifying trajectory analysis and speed calculations. This high-throughput video tracking method (i.e., a method capable of rapidly processing large amounts of movement data) greatly reduced the labor of manual observation and provided complete, objective time-series data [9, 16]. An overview of the experimental design and data analysis pipeline is provided in Fig. 1. All experiments were performed in accordance with standard *Daphnia* acute toxicity test guidelines (NIEA B901.14B, Taiwan EPA).

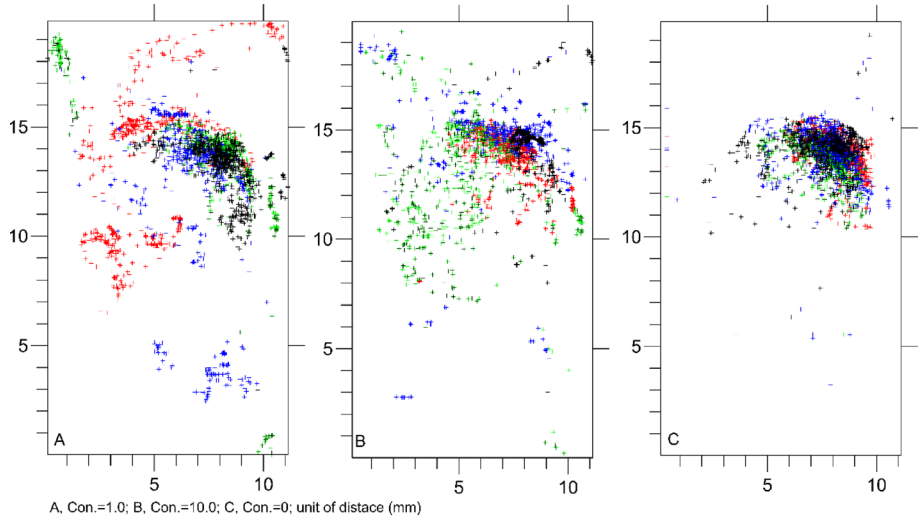


Fig. 1 Scatter plots for the trajectories of individual Daphnia

2.3 Data preprocessing and deep learning model

After obtaining the raw trajectory data, we processed and analyzed them as follows. Based on prior literature, *Daphnia* swimming speed is a particularly sensitive metric for toxicity, while parameters like acceleration and turning behavior can also reflect stress responses [9, 15]. Therefore, we extracted seven features from each recorded motion time-series: time (t), the x and y coordinates, speed (v), acceleration (a), kinetic energy (p, used here as a proxy for momentum), and angular velocity (ω). If any abnormal values were encountered (for example, if the tracking temporarily lost the organism), we applied linear interpolation to fill in those data points.

All the collected data were compiled into a single dataset, with each sample labeled by its salinity concentration (0%, 0.01%, 0.1%, 1%, or 10%). We then randomly split the dataset into a training set (80% of the samples) and a test set (20%). After splitting, all feature values were normalized via Z-score standardization (mean set to 0 and standard deviation to 1) to eliminate differences in scale between variables and to facilitate efficient model training.

For the classification task, we selected a Multi-Layer Perceptron (MLP) neural network model. An MLP was chosen because it can capture non-linear relationships in behavioral data and has been used successfully in similar environmental modeling contexts [14, 17]. The MLP was implemented in Python (TensorFlow/Keras library). After preliminary trials, we adopted a network architecture with two hidden layers, as additional layers did not appreciably improve performance. The final model architecture consisted of an input layer with 7 neurons (one for each feature), two hidden layers with 64 and 32 neurons respectively (each using ReLU activation, and with a 50% dropout applied after the first hidden layer to mitigate overfitting), and an output layer with 5 neurons (softmax activation) corresponding to the five salinity categories. We trained the MLP using the Adam optimizer and categorical cross-entropy loss function. Training was run for 100 epochs with a batch size of 16. During training, 20% of the training data were reserved as a validation set to monitor performance and prevent overfitting. All code for model training and evaluation was written in Python using standard libraries (pandas, scikit-learn, TensorFlow, etc.).

After training, we evaluated the model on the held-out test set. We calculated the overall classification accuracy as well as the precision, recall, and F1-score for each salinity class to assess performance. Precision is defined as TP/(TP + FP), recall as TP/(TP + FN), and F1-score as the harmonic mean of precision and recall, 2·(Precision·Recall)/(Precision + Recall). We also generated a confusion matrix to identify the patterns of misclassification between classes. To facilitate reproducibility, we recorded the model's training history (loss and accuracy over epochs) and checked that the validation loss/accuracy curves had stabilized without severe overfitting. In a practical deployment, all of the above analysis steps could be automated in a user-friendly application—for instance, a participant could upload a smartphone video and the system would automatically process the data and return a water quality classification result (Figs. 2, 3, 4, 5 and 6).

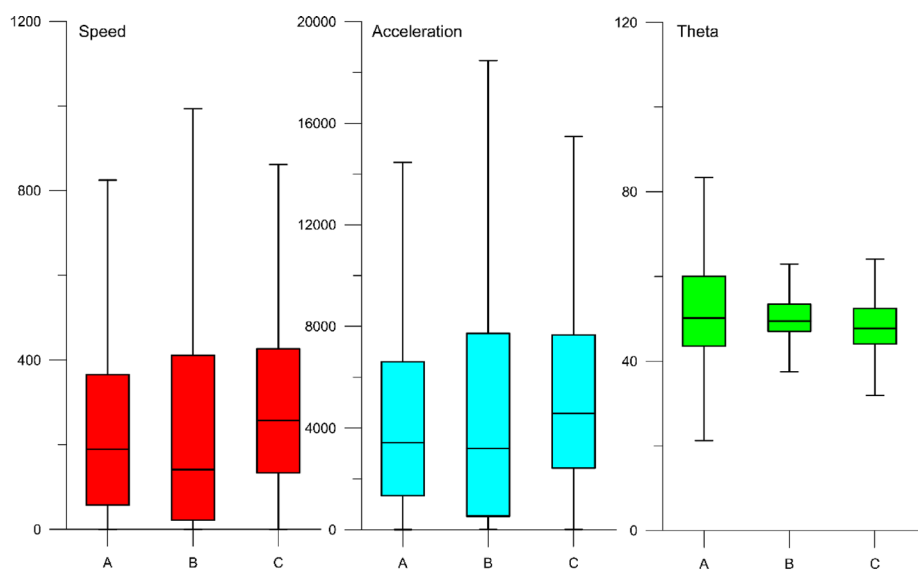
2.4 Statistical analysis

In addition to the machine learning analysis, we performed conventional statistical tests on the behavioral data. We used one-way analysis of variance (ANOVA) to determine whether *Daphnia* movement parameters differed significantly across the five salinity treatments. When ANOVA indicated a significant effect ($p < 0.05$), we conducted post-hoc comparisons (Tukey's HSD test) to identify which specific salinity groups differed from each other. We also computed Spearman's rank correlation coefficients to assess the monotonic relationship between salinity (treated as an ordinal variable) and each behavioral metric (e.g., mean speed) across samples. These statistical analyses were carried out using SPSS (version 26.0).

3 Results

3.1 Daphnia behavioral responses under different salinities

Daphnia magna showed clear changes in swimming behavior with increasing salinity. In the highest salinity treatment (10% NaCl), animals swam markedly more slowly than in



A (Conc. =1) ; B(Conc. = 10) ; C(Conc. = 0)

Fig. 2 The boxplots for distinct salinity concentrations

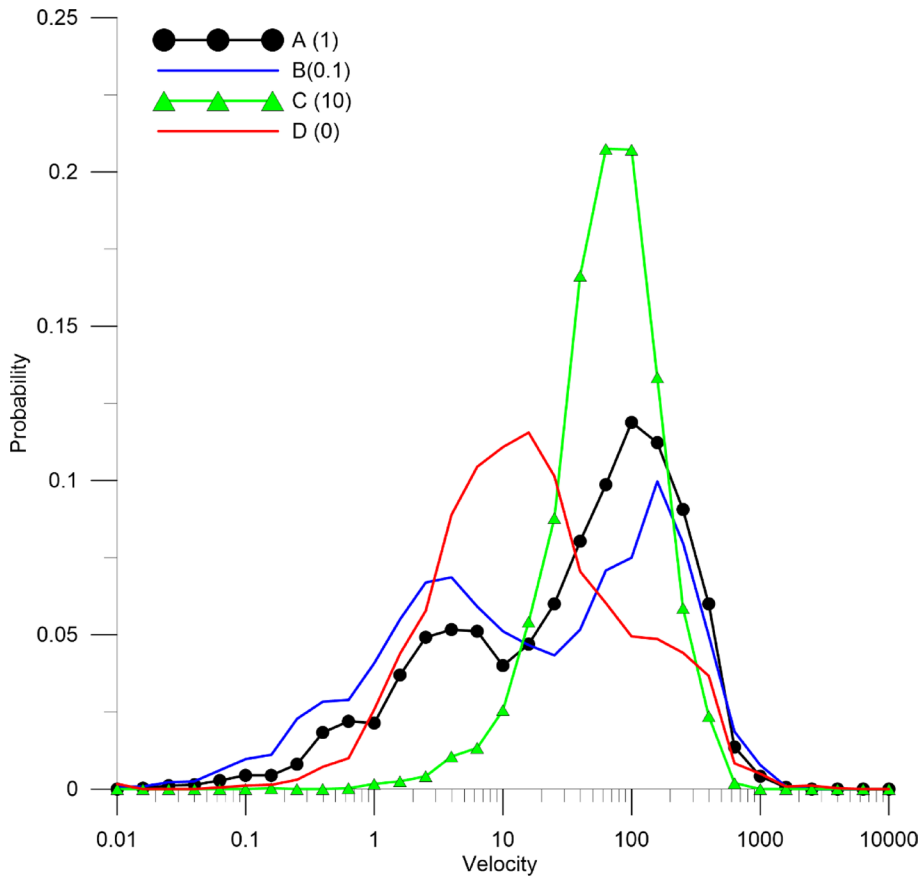


Fig. 3 The probability distribution of velocity for distinct salinity concentrations

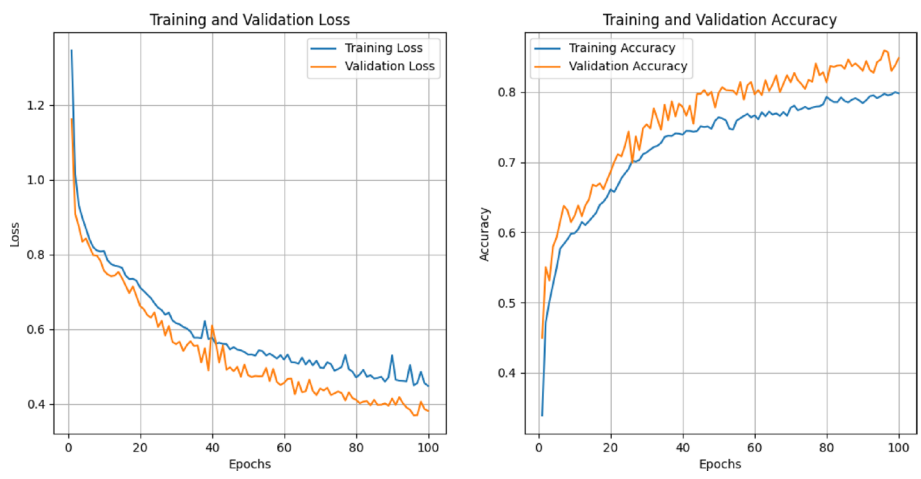


Fig. 4 The loss and accuracy of model training and validation

the control (0% NaCl). Quantitatively, the average swimming speed at 10% salinity was roughly 50% lower than at 0% (ANOVA, $p < 0.001$). For example, considering median values, the median speed in 10% salinity was about half of that in the control group. Similarly, the *Daphnia* in high salinity had reduced acceleration: the mean acceleration at 10% was approximately 30% lower than at 0% ($p < 0.01$). Some differences were also observed in movement directionality – for instance, the distribution of turning angles

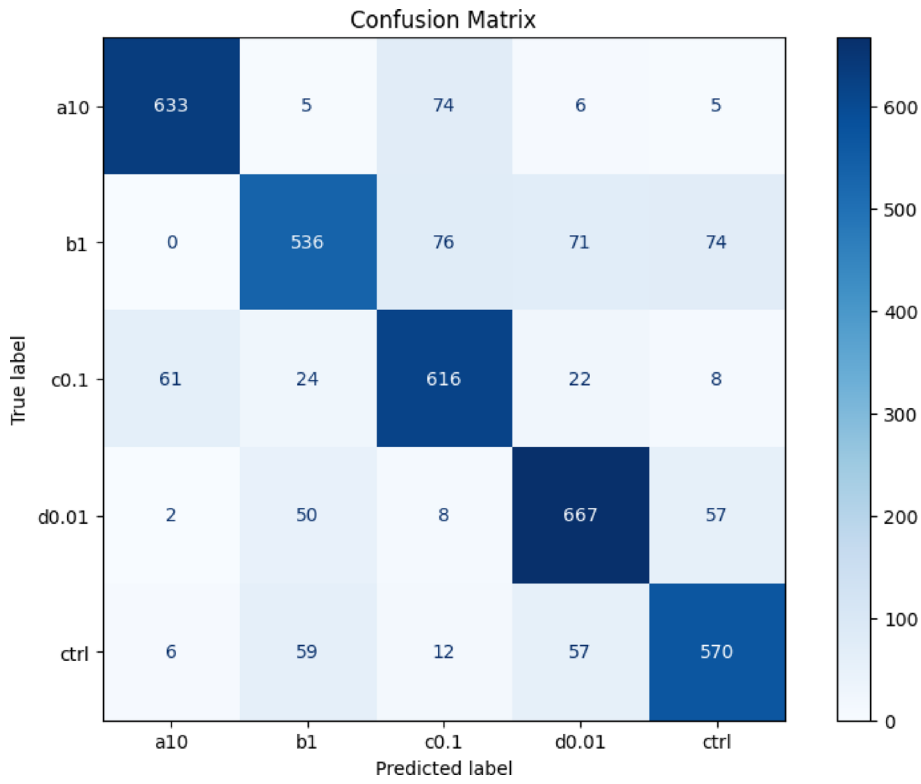


Fig. 5 The confusion matrix for distinct salinity concentrations

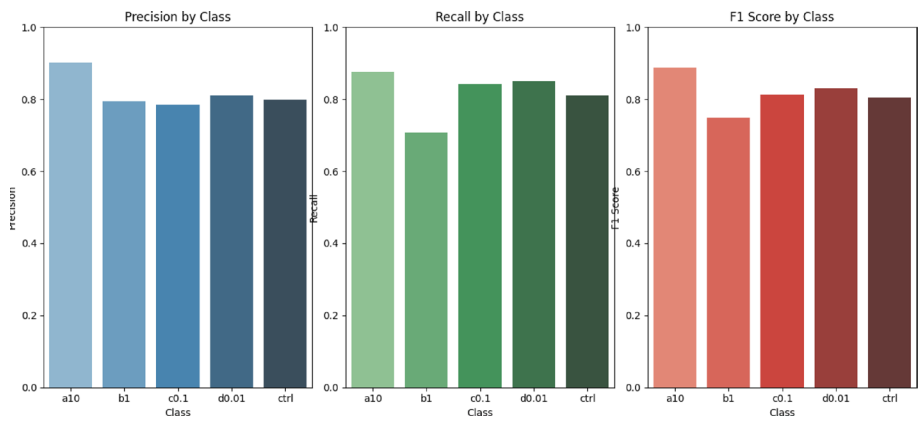


Fig. 6 The performance indicators for distinct salinity concentrations

(indicative of erratic movement) showed slight shifts between treatments (this effect was statistically significant but relatively small, $p < 0.05$). Overall, *Daphnia* exposed to greater salt concentrations, exhibited slower and less vigorous movement, indicating a strong sub-lethal stress response even at the 48-h exposure mark.

Despite the significant mean differences described above, there was considerable variability in individual behavior within each treatment. In fact, the standard deviations of speed in each salinity group were quite high. This high within-group variability means that a single metric (such as mean speed alone) did not perfectly discriminate all salinity levels. Consistent with this, a simple correlation between salinity concentration and individual *Daphnia*'s mean speed was weak (Spearman's $r \approx -0.07$, $p > 0.5$), suggesting

that relying on any one behavioral parameter may be insufficient. These results underscore the need to consider multiple features or more sophisticated analytical approaches to reliably distinguish water quality conditions based on *Daphnia* behavior.

3.2 MLP model performance

The artificial neural network model (MLP) trained on *Daphnia* behavioral features was able to classify water quality (salinity level) with good accuracy. The overall accuracy on the test set was 80%, confirming that the model captured meaningful patterns in the behavioral data. Performance metrics for each class revealed some patterns. The model was most confident and accurate in identifying the extreme conditions: both the no-stress control (0% salinity) and the highest stress (10% salinity) were classified with high precision and recall. For example, the 0% and 10% classes each achieved F1-scores around 0.85–0.89, indicating that almost all of those cases were correctly recognized. In contrast, the model found it more challenging to distinguish the intermediate salinity levels, especially the 1% salinity condition. The recall for the 1% class was the lowest among the categories (approximately 0.66), meaning that a number of 1% samples were misidentified as other similar levels. Misclassifications tended to occur between adjacent or similar concentration levels (e.g., confusing 1% with 0.1% or 10%). Nevertheless, for the three lower salinity classes (0%, 0.01%, 0.1%) precision and recall were generally in the 0.75–0.87 range, and for the highest class (10%) these metrics were around 0.88. These results indicate that the trained MLP model can reliably detect clear differences in water quality (such as pristine vs highly polluted conditions) and to a slightly lesser extent can differentiate more subtle differences (moderate salinity levels) in *Daphnia* behavioral response.

4 Discussion

The behavioral changes observed in *Daphnia magna* under salinity stress are in line with known patterns from ecotoxicology. In our experiments, *Daphnia* exposed to higher salinities swam more slowly and with lower acceleration, which is consistent with the findings of other studies that have reported reduced mobility of *Daphnia* under various toxicant exposures [9, 12]. Such reductions in swimming speed and overall activity likely reflect the physiological stress imposed by the pollutant—in this case, osmotic stress from elevated salinity—and indicate that even sub-lethal pollutant levels can impair *Daphnia*'s normal behavior. These results reinforce the idea that behavioral endpoints (like swimming speed) are sensitive indicators of environmental stress, often manifesting before more overt signs such as mortality occur [9].

It is important to note, however, the considerable individual variability we found in *Daphnia* behavior. Even within the same treatment group, some individuals were more active than others, leading to overlapping behavior distributions among different salinity levels. This variability helps explain why the simple correlation between salinity and mean speed was very weak and not significant. A single behavioral metric on its own may not always provide a clear-cut indication of water quality, especially when natural variation is high or stress effects are mild. Therefore, leveraging multiple behavioral features and analyzing them in combination is advantageous. Our approach of using a machine learning model capitalizes on this idea by effectively integrating several

behavioral parameters to discern patterns that might be missed when examining one variable at a time.

The successful application of an MLP classifier in this study demonstrates the potential of machine learning to interpret complex biological response patterns for environmental monitoring. Our model achieved about 80% accuracy in classifying five water quality levels, which is quite promising given the short exposure duration and relatively small dataset. This performance is comparable to, albeit slightly lower than, results from studies using much larger or longer-term datasets. For instance, Jeong et al. [10] applied a neural network to eight years of *Daphnia* early warning system data (with *Daphnia* continually exposed to changing conditions) and reported roughly 88% accuracy in predicting water quality categories. Considering that our data were from acute 48-h exposures and used only simple summary features of behavior, the model's performance is encouraging. It suggests that even brief observations of *Daphnia* activity contain sufficient information to gauge water condition when analyzed with an appropriate algorithm.

A variety of analytical approaches have been explored in related contexts, and our study adds evidence to the efficacy of using a straightforward neural network. Other researchers have used different machine learning techniques and confirmed their utility in analyzing *Daphnia* or similar bioindicator behaviors. For example, random forest classifiers and other algorithms have been employed to distinguish *Daphnia* motion patterns under different treatments [16], Maganathan 2020). Deep learning methods beyond MLP are also emerging in this field: Cheng et al. [18] recently used a deep neural network to identify specific chemical toxicants based on *Daphnia* locomotion tracks, and Lopes et al. [19] demonstrated that detailed trajectory analysis of *Daphnia* swimming can serve as an effective tool for environmental impact assessment. In comparison to these studies, our approach is notable for its simplicity and accessibility. We opted for a relatively compact MLP model, which was sufficient for our classification task and can be run on ordinary computing devices.

A key novelty of this work is that it shows such analyses can be performed with minimal specialized equipment. Unlike previous experimental setups that might require sophisticated imaging systems or controlled lab environments, we used just a smartphone camera and basic lighting to record *Daphnia* behavior. The ability to obtain meaningful water quality indications from consumer-grade devices is a significant step toward democratizing environmental monitoring. It implies that the barrier to participation in scientific data collection can be lowered considerably. Our results confirm that a simple video, when coupled with the appropriate analysis, can yield insights comparable to more advanced setups. This demonstrates a new paradigm for biomonitoring: one that emphasizes low-cost, real-time analysis without sacrificing scientific rigor.

The practical implications of this study are considerable, particularly for citizen science and environmental education. The method we developed could be readily employed in educational programs or community-based monitoring campaigns. For instance, students or volunteer groups could expose *Daphnia* to local water samples, record short videos of their behavior, and use an app or web tool (powered by our trained model) to assess the water's quality. This hands-on approach can enhance learning by allowing participants to directly observe a biological response to pollution and immediately quantify it. Engaging non-experts in this way not only provides more data for scientists and water managers but also raises public awareness about water quality issues. Prior work

has shown that involving students in toxicity testing with organisms like *Daphnia* can significantly improve their understanding of pollution and its effects [20]. By integrating an easy-to-use ML-based analysis, we can further increase the appeal and impact of such activities, as participants get instant feedback and a sense of contributing to a scientific assessment.

Beyond education, our approach could serve as a community-level early warning system for water quality. In practice, if many citizens each monitor their local streams or ponds with *Daphnia* tests, any significant toxic event (for example, a chemical spill leading to acute toxicity) might be detected through anomalous *Daphnia* behavior flagged by the model. This distributed monitoring could complement traditional water quality management by providing rapid, on-site biological feedback. It is analogous to existing biological early warning systems used by some water utilities, where continuous monitoring of aquatic organisms can signal contamination events [8, 10]. The difference is that our system is scalable to many users and locations, thanks to its simplicity. If implemented, environmental agencies could receive timely alerts from citizen observers, prompting more detailed investigations or interventions. In summary, by empowering the public to both learn from and contribute to water quality monitoring, our method can improve not only data coverage but also community engagement in water resource protection.

There are, of course, limitations to this study and areas for future improvement. First, we tested only a single stressor (salinity) in a controlled lab setting. Real-world water pollution is often due to complex mixtures of chemicals, and factors like temperature, pH, or the presence of predators could influence *Daphnia* behavior. Thus, one limitation is the question of generalizability: the model trained on salinity-induced behavior changes may need retraining or adjustment to accurately detect other types of pollutants. Future research should evaluate the approach with other common contaminants (e.g., heavy metals, pesticides, organic pollutants) and ideally with field water samples to verify that the system can differentiate clean versus polluted natural waters. Second, our study focused on one species, *Daphnia magna*, as a representative bioindicator. While *D. magna* is widely used and very sensitive, different species (including other *Daphnia* or even different genera) might respond differently to certain stressors. Expanding the approach to incorporate multiple bioindicator species or strains could broaden the applicability of the method, but would also require additional model training to accommodate species-specific behaviors. Third, the MLP model we employed was relatively simple. It performed well for our purposes, but more advanced machine learning models (such as convolutional neural networks or recurrent neural networks that can directly analyze video frames or time-series data) could potentially capture subtler aspects of the behavior. Implementing more complex models might further improve accuracy or enable the detection of more nuanced patterns (for example, subtle changes in swimming trajectory shapes). However, any increase in model complexity should be balanced with the need to maintain the approach's accessibility and real-time capabilities. Finally, from a user perspective, the current process still involves some technical steps (video recording, data processing, running the model) that might be challenging for laypersons without guidance. Developing an integrated software tool or smartphone application is a future step that would streamline these steps – essentially hiding the technical complexity behind a user-friendly interface. With such improvements, we believe this

Daphnia-ML framework can move from a proof-of-concept in the lab toward a practical tool for widespread use in water quality monitoring.

5 Conclusion

In summary, this study presents a novel method for water quality assessment that leverages a biological indicator and artificial intelligence. We showed that sub-lethal pollution exposure (salinity stress) leads to quantifiable changes in *Daphnia magna* swimming behavior, and that a simple machine learning model (MLP) can automatically classify different water quality levels based on those behavioral features with around 80% accuracy. These findings confirm that behavioral biomonitoring, when augmented with modern data analysis, can serve as an effective early indicator of water quality degradation. The approach we demonstrated is significant in that it bridges the gap between complex environmental data and public-friendly monitoring: using just a smartphone and an accessible algorithm, non-experts can obtain meaningful insights into water health. The method's simplicity and low cost make it especially suitable for citizen science and educational initiatives, potentially empowering communities to engage in local water resource management. We acknowledge that our study was conducted in controlled conditions with a single stress factor, so further validation is required in more complex real-world scenarios. Future work should extend this approach by testing additional pollutants, evaluating performance in field environments, and refining the machine learning model (or exploring more advanced architectures) for even greater accuracy and generality. Despite these limitations, our work provides a foundation for a new class of easy-to-use, real-time biomonitoring tools. With continued development, this *Daphnia* behavior-based ML framework could become a valuable complement to traditional water quality monitoring, offering rapid biological feedback and fostering broader public participation in environmental protection.

Authors' contributions

Conceptualization, Y.-J. Chang and T.-Y. Yu; methodology, H.-S. Chang, T.-K. Yu, and C.-H. Chang; software, Y.-J. Chang, C.-H. Chang, and T.-K. Yu; validation, Y.-J. Chang and T.-Y. Yu; data curation, H.-S. Chang and C.-H. Chang; writing—original draft, Y.-J. Chang and T.-Y. Yu; writing—review and editing, T.-K. Yu; supervision, Y.-J. Chang. All authors have read and agreed to the published version of the manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This article does not contain any studies with human participants or live vertebrates that require ethical approval. *Daphnia magna* is an invertebrate test organism commonly used in ecotoxicology and not subject to animal welfare regulations.

Consent for publication

Not applicable. (No individual person's data are included.)

Competing interests

The authors declare no competing interests.

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